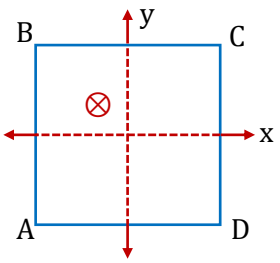
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Faraday's Law and Lenz's Law

1. A square coil *ABCD* is placed in *x-y* plane with its centre at origin. A long straight wire, passing through origin, carries a current in negative *z*-direction. Current in this wire increases with time. The induced current in the coil is :



(A) clockwise (B) anticlockwise (C) zero (D) alternating

PEM045

2. Two identical coaxial circular loops carry a current *i* each circulating in the same direction. If the loops approach each other

(A) the current in each will decrease

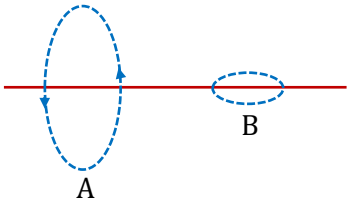
(B) the current in each will increase

(C) the current in each will remain the same

(D) the current in one will increase and in other will decrease

PEM046

3. In the given figure the centre of a small conducting circular loop *B* lies on the axis of bigger circular loop *A* and their axis are mutually perpendicular. An anticlockwise (when viewed from the side of *B*) current in the loop *A* start increasing then :-



(A) current induced in the loop *B* is in clockwise direction (when viewed from above the *B*).

(B) current induced in the loop *B* is in anti-clockwise direction (when viewed from above the *B*).

(C) current must induced in the loop *B* but its direction can not be predicted.

(D) no current is induced in the loop *B*.

PEM048

For More Join: @IITJEE_Advanced

4. A vertical bar magnet is dropped from position on the axis of a fixed metallic coil as shown in fig-I. In fig. II the magnet is fixed and horizontal coil is dropped. The acceleration of the magnet and coil are a_1 and a_2 respectively then :-

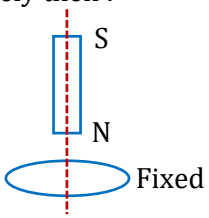


Fig. (I)

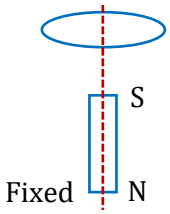
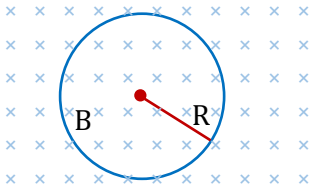


Fig. (II)

(A) $a_1 > g$, $a_2 > g$ (B) $a_1 > g$, $a_2 < g$ (C) $a_1 < g$, $a_2 < g$ (D) $a_1 < g$, $a_2 > g$

PEM049

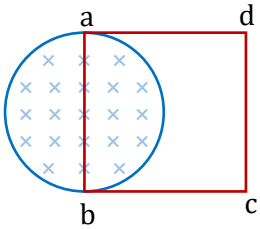
5. A conducting loop of radius R is present in a uniform magnetic field B perpendicular to the plane of the ring. If radius R varies as a function of time 't', as $R = R_0 + t$. The e.m.f induced in the loop is



(A) $2\pi(R_0 + t)B$ clockwise (B) $\pi(R_0 + t)B$ clockwise
(C) $2\pi(R_0 + t)B$ anticlockwise (D) zero

PEM052

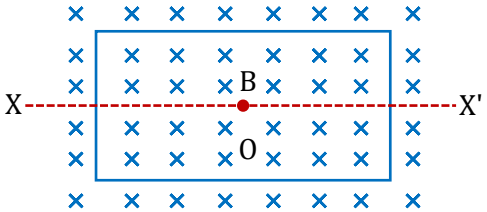
6. A uniform magnetic field B exists in a cylindrical region of radius 10 cm as shown in figure. A uniform wire of length 80 cm and resistance 40 ohms is bent into a square frame and is placed with one side along a diameter of cylindrical region. If the magnetic field increases at a constant rate 0.1 tesla per second, find the current induced in the frame.



(A) $3.14 \times 10^{-5}\ A$ (B) $3.9 \times 10^{-5}\ A$ (C) $1 \times 10^{-5}\ A$ (D) $2.8 \times 10^{-5}\ A$

PEM053

7. A rectangular coil is placed in a region having a uniform magnetic field B perpendicular to the plane of the coil. An emf will not be induced in the coil if the



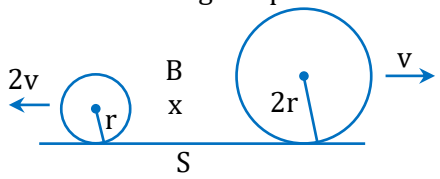
(A) magnetic field is increased uniformly
(B) magnetic field is switched off
(C) coil is rotated about the axis XX'
(D) coil is rotated about an axis perpendicular to the plane of the coil and passing through its centre O

PEM117

For More Join: @IITJEE_Advanced

Motional EMF

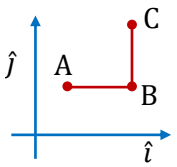
8. Two conducting rings of radii r and $2r$ move in opposite directions with velocities $2v$ and v respectively on a conducting surface S. There is a uniform magnetic field of magnitude B perpendicular to the plane rings. The potential difference between the highest points of the two rings is:



(A) zero (B) $2rvB$ (C) $4rvB$ (D) $8rvB$

PEM121

9. There is a uniform magnetic field B normal to the xy plane. A conductor ABC has length $AB = \ell_1$, parallel to the x-axis, and length $BC = \ell_2$, parallel to the y-axis. ABC moves in the xy plane with velocity $v_x\hat{\imath} + v_y\hat{\jmath}$. The potential difference between A and C is proportional to :-

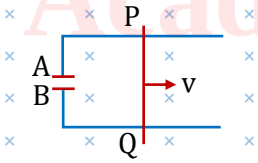


(A) $v_x\ell_1 + v_y\ell_2$ (B) $v_x\ell_2 + v_y\ell_1$ (C) $v_x\ell_2 - v_y\ell_1$ (D) $v_x\ell_1 - v_y\ell_2$

PEM055

10. A conducting rod PQ of length $L = 1.0\ m$ is moving with a uniform speed $v = 20\ m/s$ in a uniform magnetic field $B = 4.0\ T$ directed into the paper. A capacitor of capacity $C = 10\ \mu F$ is connected as shown in figure. Then

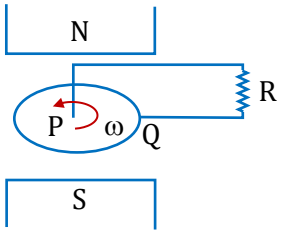
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(A) $q_A = +\ 800\mu C$ and $q_B = -\ 800\mu C$
(B) $q_A = -\ 800\mu C$ and $q_B = +\ 800\mu C$
(C) $q_A = 0 = q_B$
(D) charge stored in the capacitor increases exponentially with time.

PEM056

11. A metal disc rotates freely, between the poles of a magnet in the direction indicated. Brushes P and Q make contact with the edge of the disc and the metal axle. What current, if any, flows through R?



(A) a current from P to Q clockwise
(B) a current from Q to P clockwise
(C) no current, because the emf in the disc is opposed by the back emf
(D) no current, because the emf induced in one side of the disc is opposed by the emf induced in the other side.

PEM063

For More Join: @IITJEE_Advanced

12. A metallic rod of length ℓ is rotated in a horizontal plane with angular velocity ω where no magnetic field exists. The EMF induced in the rod is (Mass of e^- is m and charge of electron is e)



(A) $\frac{m\ell\omega^2}{e}$ (B) $\frac{m\omega\ell^2}{2e}$ (C) $\frac{m\omega^2\ell^2}{2e}$ (D) $\frac{m\omega^2\ell^2}{e}$

PEM064

13. An electron is moving in a circular orbit of radius R with an angular acceleration α. At the centre of the orbit is kept a conducting loop of radius r, ($r << R$). The e.m.f induced in the smaller loop due to the motion of the electron is :-

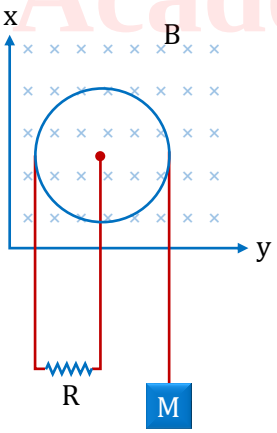
(A) zero, since charge on electron in constant (B) $\frac{\mu_0 er^2}{4R}\alpha$

(C) $\frac{\mu_0 er^2}{4\pi R}\alpha$ (D) none of these

PEM065

14. The block of mass (M) is connected by thread, which is wound on a pulley, free to rotate about fixed horizontal axis as shown. A uniform magnetic field B exists in a horizontal plane. The disc is connected with the resistance R as shown. Calculate the terminal velocity of the block if it was released from rest. Treat pulley as uniform metallic disc of radius L.

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(A) $\frac{4mgR}{B^2L^2}$ (B) $\frac{3mgR}{4B^2L^2}$ (C) $\frac{2mgR}{B^2L^2}$ (D) $\frac{3mgR}{2B^2L^2}$

PEM068

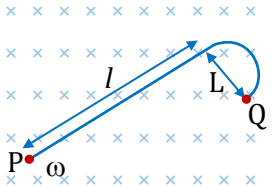
15. The magnetic field in a region is given by $\vec{B} = B_0\left(1+\frac{x}{a}\right)\hat{k}$. A square loop of edge - length d is placed with its edge along x and y axis. The loop is moved with constant velocity $\vec{V} = V_0\hat{\imath}$. The emf induced in the loop is

(A) $\frac{V_0B_0d^2}{a}$ (B) $\frac{V_0B_0d^2}{2a}$ (C) $\frac{V_0B_0a^2}{d}$ (D) None

PEM069

For More Join: @IITJEE_Advanced

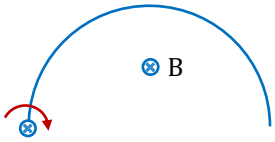
16. When a '*J*' shaped conducting rod is rotating in its own plane with constant angular velocity ω, about one of its end *P*, in a uniform magnetic field $\vec{B}$ directed normally into the plane of paper then magnitude of emf induced across it will be



(A) $B\omega\sqrt{L^2+l^2}$ (B) $\frac{1}{2}B\omega L^2$ (C) $\frac{1}{2}B\omega(L^2+l^2)$ (D) $\frac{1}{2}B\omega l^2$

PEM070

17. A semicircular wire of radius *R* is rotated with constant angular velocity ω about an axis passing through one end and perpendicular to the plane of the wire. There is a uniform magnetic field of strength *B*. The induced e.m.f. between the ends is–

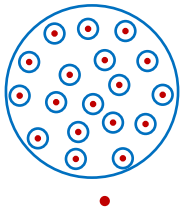


(A) $B\omega R^2/2$ (B) $2B\omega R^2$
(C) is variable (D) none of these

PEM123

Induced Electric Field

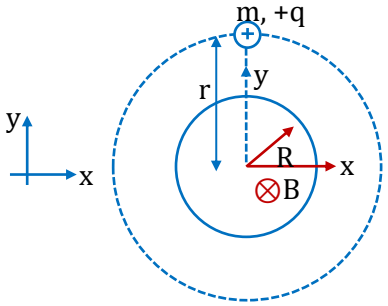
18. As shown, a uniform magnetic field *B* pointing out of the paper plane is confined in the shaded area of radius *r*. At a distance *R* (*R* > *r*) from the center of the shaded area there is a point particle of mass *m* and carrying charge *q*. Speed of the particle if the magnetic field in the region is quickly changed to zero will be.



(A) $v=\frac{qBr^2}{2mR}$ (B) 0 (C) $v=\frac{qBr^2}{mR}$ (D) $v=\frac{2qBr^2}{mR}$

PEM073

19. Circular region of radius *R* has uniform magnetic field $\vec{B} = B_0 + B_0t(-\hat{k})$. At $t = 0$ acceleration of charged particle :-



(A) $\frac{qB_0R^2}{2mr}$ (B) $\frac{qB_0R}{2mr}$ (C) $\frac{qB_0R^3}{2mr^2}$ (D) $\frac{qB_0R^2}{mr}$

PEM075

For More Join: @IITJEE_Advanced

20. Given a ring of mass M, radius r resting on a rough horizontal plane, with coefficient of friction 'μ'. A cylindrical, vertical magnetic field passing through ring is switched on at $t = 0$ which varies with time as $B = Kt^2$. Ring has charge Q distributed uniformly over it. Find the time after which ring starts rotating.

(A) $\frac{\mu Mg}{QKr}$ (B) $\frac{2\mu Mg}{QKr}$ (C) $\frac{\mu Mg}{2QKr}$ (D) $\frac{\mu Mg}{4QKr}$

PEM076

21. **Statement–1 :** For a charged particle moving from point P to point Q the net work done by an induced electric field on the particle is independent of the path connecting point P to point Q.

Statement–2 : The net work done by a conservative force on an object moving along closed loop is zero.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

PEM080

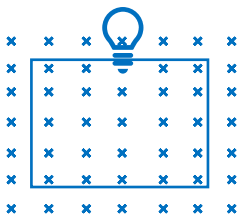
22. There exists a uniform cylindrically symmetric magnetic field directed along the axis of a cylinder but varying with time as $B = kt$. If an electron is released from rest in this field at a distance of 'r' from the axis of cylinder, its acceleration, just after it is released would be (e and m are the electronic charge and mass respectively)

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(A) $\frac{erk}{2m}$ (B) $\frac{erk}{m}$ (C) $\frac{erk}{4m}$ (D) $\frac{2erk}{m}$

PEM081

23. A square wire loop of 10.0 cm side lies at right angles to a uniform magnetic field of 20T. A 10 V light bulb is in a series with the loop as shown in the fig. The magnetic field is decreasing steadily to zero over a time interval Δt. The bulb will shine with full brightness if Δt is equal to :-



(A) 20 ms (B) 0.02 ms (C) 2 ms (D) 0.2 ms

PEM125

Self-Inductance and Mutual Inductance

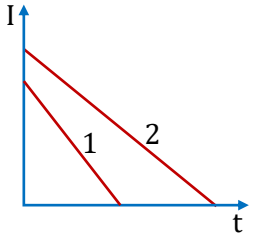
24. A current of 2A is increasing at a rate of 4 A/s through a coil of inductance 2H. The energy stored in the inductor per unit time is :-

(A) 2 J/s (B) 1 J/s (C) 16 J/s (D) 4 J/s

PEM083

For More Join: @IITJEE_Advanced

25. Two identical inductance carry currents that vary with time according to linear laws (as shown in figure). In which of two inductance is the self induction emf greater?



(A) 1 (B) 2

(C) same (D) data are insufficient to decide

PEM084

26. When the current changes from $+2A$ to $-2A$ in 0.05 second, an EMF of $8V$ is induced in the coil. The coefficients of self inductance of the coil.

(A) 0.05 H (B) 0.1 H (C) 0.2 H (D) 0.8 H

PEM087

27. A small coil of radius r is placed at the centre of a large coil of radius R, where $R >> r$. The coils are coplanar. The coefficient of mutual inductance between the coils is :-

(A) $\frac{\mu_0 \pi r}{2R}$ (B) $\frac{\mu_0 \pi r^2}{2R}$ (C) $\frac{\mu_0 \pi r^2}{2R^2}$ (D) $\frac{\mu_0 \pi r}{2R^2}$

PEM088

28. A long straight wire is placed along the axis of a circular ring of radius R. The mutual inductance of this system is :-

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(A) $\frac{\mu_0 R}{2}$ (B) $\frac{\mu_0 \pi R}{2}$ (C) $\frac{\mu_0}{2}$ (D) 0

PEM089

29. There are 10 turns in coil M and 15 turns in coil N. If a current of $2A$ is passed through coil M then the flux linked with coil N is 1.8×10^{-3} Wb. If a current of $3A$ is passed through coil N then flux linked with coil M is :-

(A) 1.7×10^{-3} Wb (B) 2.7×10^{-3} Wb (C) 1.7×10^{-5} Wb (D) 2.7×10^{-5} Wb

PEM090

30. If the difference between the equivalent inductance in the following figures is nL then find the value of n. Given coupling coefficient is $C = \sqrt{2}$ (Where coupling coefficient is defined as $C = \frac{\sqrt{L_1 L_2}}{M}$)

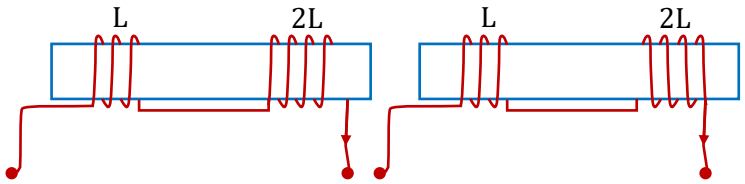


Figure (A) **Figure (B)**

(A) 2 (B) 3 (C) 4 (D) 5

PEM097

For More Join: @IITJEE_Advanced

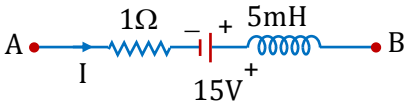
31. Two coaxial solenoids are made by winding thin insulated wire over a pipe of cross-sectional area $A = 10\ cm^2$ and length = 20 *cm*. If one of the solenoids has 300 turns and the other 400 turns, their mutual inductance is $\left(\mu_0 = 4\pi \times 10^{-7}\ TmA^{-1}\right)$

(A) $2.4\pi \times 10^{-5}\ H$ (B) $4.8\ \pi \times 10^{-4} H$ (C) $4.8\ \pi \times 10^{-5} H$ (D) $2.4\ \pi \times 10^{-4} H$

PEM128

R–L Circuit

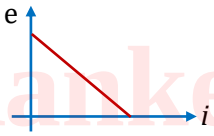
32. The network shown in the figure is part of a complete circuit. If at a certain instant, the current I is $5A$ and it is decreasing at a rate of $10^3\ As^{-1}$ then $V_B - V_A$ equals

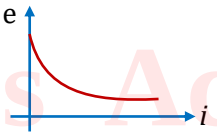


(A) 20 *V* (B) 15 *V* (C) 10 *V* (D) 5 *V*

PEM086

33. If an *L-R* circuit is connected to a battery of constant e.m.f. E switch S is closed at time $t = 0$. If e denotes the magnitude of induced e.m.f. across inductor and i is the current in the circuit at any time t. Then which of the following graphs shows the variation of e with i ?

(A) 

(B) 

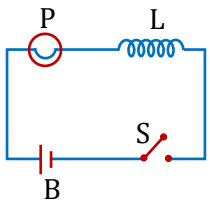
(C) 

(D) 

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PEM098

34. In figure, a lamp P is in series with an iron-core inductor L. When the switch S is closed, the brightness of the lamp rises relatively slowly to its full brightness than it would do without the inductor. This is due to



(A) the low resistance of P

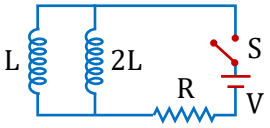
(B) the induced-emf in L

(C) the low resistance of L

(D) the high voltage of the battery B

PEM099

35. In the circuit shown, the switch is closed at $t = 0$. The current through the inductor of inductance L varies with time as :



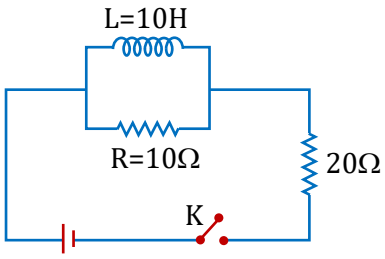
(A) $(2V/3R)\ [1 - \exp(-3Rt/2L)]$ (B) $(2V/3R)\ \exp(-3Rt/2L)$

(C) $(V/R)\ [1 - \exp(-3Rt/2L)]$ (D) $(V/R)\ [1 - \exp(-Rt/2L)]$

PEM100

For More Join: @IITJEE_Advanced

36. Two resistors of 10Ω and 20Ω and an ideal inductor of $10H$ are connected to a $2V$ battery as shown. The key K is shorted at time $t = 0$. Find the initial ($t = 0$) and final ($t \to \infty$) currents through battery.



(A) $\frac{1}{15}A, \frac{1}{10}A$ (B) $\frac{1}{10}A, \frac{1}{15}A$ (C) $\frac{2}{15}A, \frac{1}{10}A$ (D) $\frac{1}{15}A, \frac{2}{25}A$

PEM101

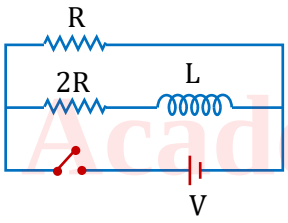
37. An LR circuit with a battery is connected at $t = 0$. Which of the following quantities is not zero just after the circuit is closed ?

(A) current in the circuit (B) magnetic field energy in the inductor

(C) power delivered by the battery (D) emf induced in the inductor

PEM103

38. Find the ratio of time constant in build up and decay in the circuit as shown in figure :-

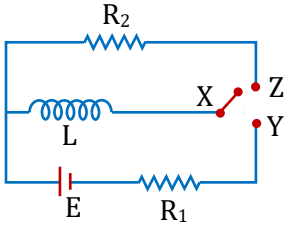


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(A) 1 : 1 (B) 3 : 2 (C) 2 : 3 (D) 1 : 3

PEM106

39. In the circuit shown, X is joined to Y for a long time, and then X is joined to Z. The total heat produced in R_2 is :



(A) $\frac{LE^2}{2R_1^2}$ (B) $\frac{LE^2}{2R_2^2}$ (C) $\frac{LE^2}{2R_1R_2}$ (D) $\frac{LE^2R_2}{2R_1^2}$

PEM107

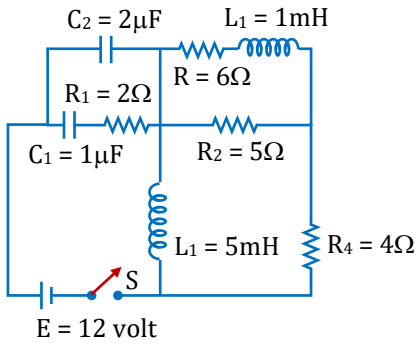
40. In a L–R decay circuit, the initial current at $t = 0$ is I. The total charge that has flown through the resistor till the energy in the inductor has reduced to one–fourth of its initial value, is

(A) LI/R (B) $LI/2R$ (C) $LI\sqrt{2}/R$ (D) None

PEM108

For More Join: @IITJEE_Advanced

41. Find current through the battery just after switch S is closed. Initially all the capacitors are uncharged:



(A) $\frac{4}{3}A$ (B) $\frac{5}{3}A$ (C) 0 A (D) $\frac{3}{5}A$

PEM129

L–C Oscillations

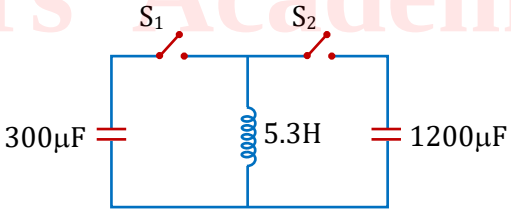
42. The inductor in a L–C oscillation has a maximum potential difference of 16 V and maximum energy of 640 μJ. Find the value of capacitor in μF in L–C circuit.

(A) 5 (B) 4 (C) 3 (D) 2

PEM113

43. The 300 μF capacitor in figure is initially charged to 16 V, the 1200 μF capacitor is uncharged and the switches are both open. What is the maximum voltage (in V) to which you can charge the 1200 μF capacitor by the proper closing and opening of the two switches.

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(A) 2 (B) 8 (C) 5 (D) 9

PEM115

44. A condenser of capacity 6 μF is fully charged using a 6-volt battery. The battery is removed and a resistanceless 0.2 mH inductor is connected across the condenser. The current which is flowing through the inductor when one-third of the total energy is in the magnetic field of the inductor is :-

(A) 0.1 A (B) 0.2 A (C) 0.4 A (D) 0.6 A

PEM130

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. A current carrying infinitely long wire is kept along the diameter of a circular wire loop, without touching it. The correct statement(s) is (are)

(A) The emf induced in the loop is zero if the current is constant.

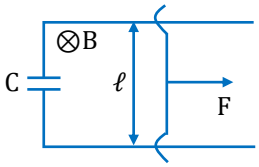
(B) The emf induced in the loop is infinite if the current is constant.

(C) The emf induced in the loop is zero if the current decreases at a steady rate.

(D) The emf induced in the loop is finite if the current decreases at a steady rate.

PEM131

2. A conducting wire of length ℓ and mass m can slide without friction on two parallel rails and is connected to capacitance C initial charge zero. Whole system lies in a magnetic field B and a constant force F is applied to the rod at $t = 0$. If initial speed of rod is zero. Then



(A) Speed of the rod at time $t = \dfrac{Ft}{B^2\ell^2C + m}$

(B) The rod moves with constant acceleration

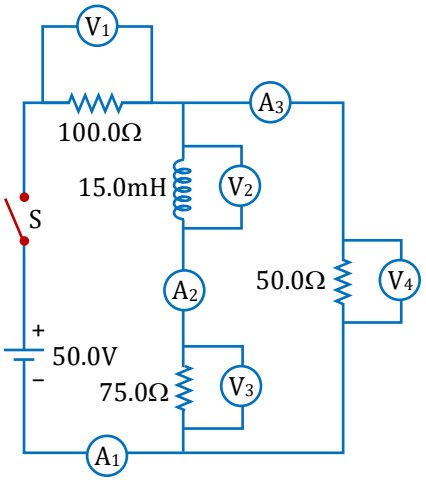
(C) The charge of capacitor varies linearly with velocity

(D) Charge on the capacitor will never attain steady state value.

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PEM132

3. In the circuit shown in figure, just after switch S is closed, the reading in each ammeter (in amp) and voltmeter (in V) will be :-



(A) $A_1 = \dfrac{1}{3}$, $A_2 = 0$ and $A_3 = \dfrac{1}{3}$

(B) $A_1 = \dfrac{5}{13}$, $A_2 = \dfrac{2}{13}$ and $A_3 = \dfrac{3}{13}$

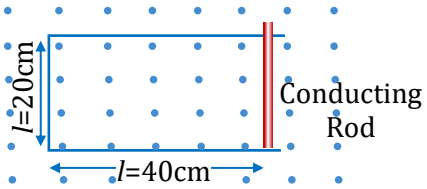
(C) $V_1 = \dfrac{100}{3}$, $V_3 = 0$

(D) $V_2 = \dfrac{50}{3}$, $V_4 = \dfrac{100}{3}$

PEM133

For More Join: @IITJEE_Advanced

4. Figure shows a conducting rod of negligible resistance that can slide on smooth U-shaped rail made of wire of resistance 1 Ω/m. Position of the conducting rod at $t = 0$ is shown. A time dependent magnetic field $B = 2t$ Tesla is switched on at $t = 0$. After the magnetic field is switched on, the conducting rod is moved to the left perpendicular to the rails at constant speed 5 cm/s by some external agent.



(A) The current in the loop at $t = 0$ due to induced emf is 0.16 A, clockwise

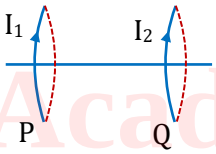
(B) At $t = 2s$, induced emf has magnitude 0.08 V

(C) The magnitude of the force required to move the conducting rod at constant speed 5 cm/s at $t = 2s$, is equal to 0.08 N

(D) The magnitude of the force required to move the conducting rod at constant speed 5 cm/s at $t = 2s$, is equal to 0.16 N

PEM024

5. Two circular coils P and Q are fixed coaxially and carry currents I_1 and I_2 respectively



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(A) if $I_2 = 0$ and P moves towards Q, a current in the same direction as I_1 is induced in Q

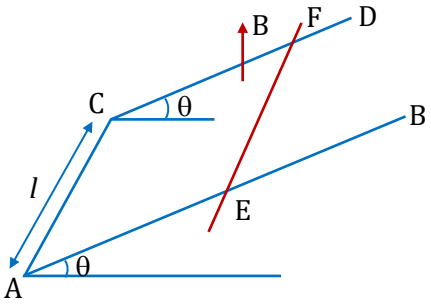
(B) if $I_1 = 0$ and Q moves towards P, a current in the opposite direction to that of I_2 is induced in P

(C) when $I_1 \neq 0$ and $I_2 \neq 0$ are in the same direction then the two coils tend to move apart

(D) when $I_1 \neq 0$ and $I_2 \neq 0$ are in opposite directions then the coils tends to move apart

PEM026

6. AB and CD are smooth parallel rails, separated by a distance l, and inclined to the horizontal at an angle θ. A uniform magnetic field of magnitude B, directed vertically upwards, exists in the region. EF is a conductor of mass m, carrying a current i. For EF to be in equilibrium,



(A) i must flow from E to F

(B) Bil = $mg \tan\theta$

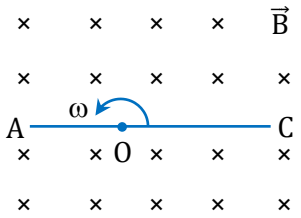
(C) Bil = $mg \sin\theta$

(D) Bil = mg

PEM060

For More Join: @IITJEE_Advanced

7. A conducting rod AC of length 4ℓ is rotated about a point O in a uniform magnetic field directed into the paper. $AO = \ell$ and $OC = 3\ell$. Then



(A) $V_A - V_0 = \frac{B\omega\ell^2}{2}$

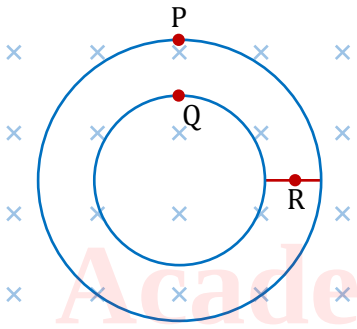
(B) $V_0 - V_C = \frac{9}{2} B\omega\ell^2$

(C) $V_A - V_C = 4B\omega\ell^2$

(D) $V_C - V_0 = \frac{9}{2} B\omega\ell^2$

PEM135

8. Figure shows plane figure made of a conductor located in a magnetic field along the inward normal to the plane of the figure. The magnetic field starts diminishing. Then the induced current



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(A) at point P is clockwise

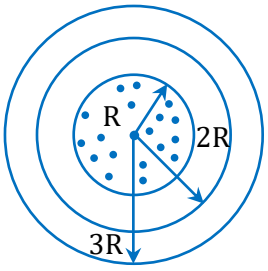
(B) at point Q is anticlockwise

(C) at point Q is clockwise

(D) at point R is zero

PEM025

9. A uniform magnetic field $\vec{B}$ exists in circular region of radius R as shown which is decreasing with time at a constant rate. There is a concentric wooden ring that has uniformly distributed charge Q and radius $2R$ and another concentric uncharged copper ring of radius $3R$, then select correct alternative. (Assuming rings are placed on frictionless surface)



(A) There is no induced current in wooden ring

(B) Induced electric field in wooden ring is more in magnitude then copper ring

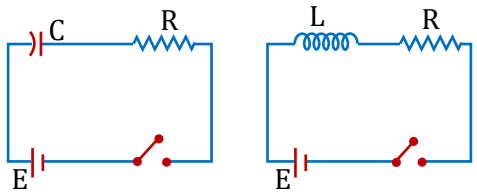
(C) Induced electric field in copper ring is more in magnitude then wooden ring

(D) Wooden ring will start rotating

PEM137

For More Join: @IITJEE_Advanced

10. The switches in figures (a) and (b) are closed at $t = 0$

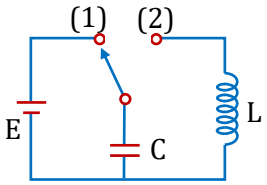


(a) **(b)**

(A) The charge on C just after $t = 0$ is EC.
(B) The charge on C long after $t = 0$ is EC.
(C) The current in L just after $t = 0$ is E/R.
(D) The current in L long after $t = 0$ is E/R.

PEM104

11. Initially key was placed on (1) till the capacitor got fully charged. Key is placed on (2) at $t = 0$. The time when the energy in both capacitor and inductor will be same-



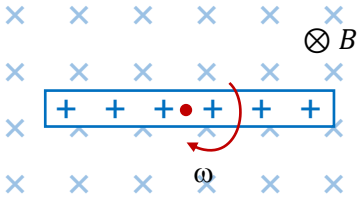
(A) $\frac{\pi\sqrt{LC}}{4}$ (B) $\frac{\pi\sqrt{LC}}{2}$ (C) $\frac{5\pi\sqrt{LC}}{4}$ (D) $\frac{5\pi\sqrt{LC}}{2}$

PEM116

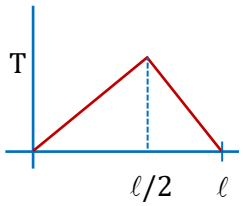
COMPREHENSION TYPE QUESTIONS

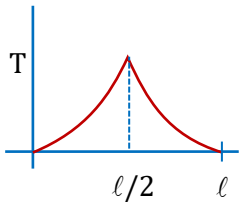
PARAGRAPH FOR QUESTION NO. 12 AND 13

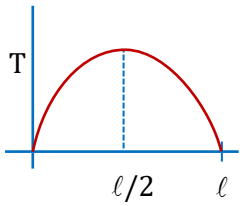
A uniform rod of mass m length l is of a non-conducting material and has a uniform charge density λ. The rod is rotated with constant ω in a gravity free region having uniform magnetic field B.

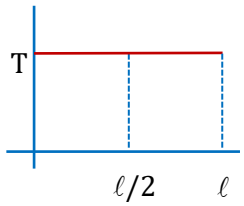


12. Taking origin at one end of rod graph of magnitude of tension with length of rod would be :-

(A) 

(B) 

(C) 

(D) 

PEM071

13. The ratio of magnitude of tension at $\frac{\ell}{2}$ and $\frac{3\ell}{4}$ distance from one end would be :-

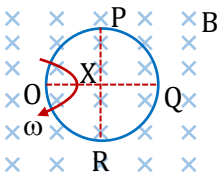
(A) 1 : 1 (B) 4 : 3 (C) 3 : 2 (D) 5 : 4

PEM072

For More Join: @IITJEE_Advanced

PARAGRAPH FOR QUESTION NO. 14 TO 16

A conducting ring of radius a is rotated about a point O on its periphery as shown in the figure in a plane perpendicular to uniform magnetic field B which exists everywhere. The rotational velocity is ω.



14. Choose the correct statement(s) related to the potential of the points P, Q and R

(A) $V_P - V_O > 0$ and $V_R - V_O < 0$ (B) $V_P = V_R < V_O$

(C) $V_O > V_P = V_Q$ (D) $V_Q - V_P = V_P - V_O$

PEM027

15. Choose the correct statement(s) related to the magnitude of potential differences

(A) $V_P - V_O = \frac{1}{2} B\omega a^2$ (B) $V_P - V_Q = \frac{1}{2} B\omega a^2$

(C) $V_Q - V_O = 2B\omega a^2$ (D) $V_P - V_R = 2B\omega a^2$

PEM028

16. Choose the correct statement(s) related to the induced current in the ring

(A) Current flows from $Q \longrightarrow P \longrightarrow O \longrightarrow R \longrightarrow Q$

(B) Current flows from $Q \longrightarrow R \longrightarrow O \longrightarrow P \longrightarrow Q$

(C) Current flows from $Q \longrightarrow P \longrightarrow O$ and from $Q \longrightarrow R \longrightarrow O$

(D) No current flows

PEM029

17. Values of equivalent resistance and equivalent inductance are respectively

(A) 1.5 Ω, 4.5×10^{-5} H (B) 3 Ω, 9×10^{-5} H

(C) 6 Ω, 1.8×10^{-4} H (D) 12 Ω, 3.6×10^{-4} H

PEM110

18. Time constant for the circuit is

(A) 15 μs (B) 45μs (C) 30μs (D) 60 μs

PEM111

19. Steady state current through the battery is

(A) 12A (B) 8A (C) 2A (D) 4A

PEM112

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For More Join: @IITJEE_Advanced

EXERCISE (S)

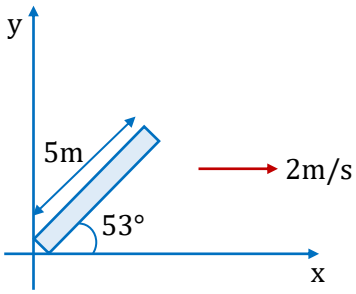
Faraday's Law and Lenz's Law

1. An air -cored solenoid of length 30 cm. area of cross-section 25 cm^2 and number of turns 500, carries a current of 2.5 A. The current is suddenly switched off in a brief time of $10^{-3}s$. How much is the average back emf induced across the ends of the open switch in the circuit? Ignore the variation in magnetic field near the ends of the solenoid.

PEM139

Motional EMF

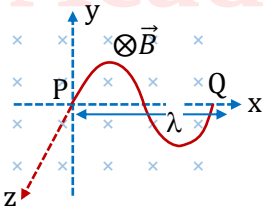
2. A conducting rod PQ of $l = 5m$ oriented as shown is moving with $V = (2m/s)\,\hat{\imath}$ without any rotation in a uniform magnetic field $(3\hat{\jmath} + 4\hat{k})$ Tesla. Emf (V) induced in the rod is



PEM043

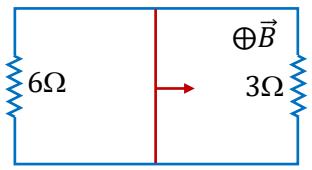
3. A wire forming one cycle of sine curve is moved in x-y plane with velocity $\vec{V} = V_x\hat{\imath} + V_y\hat{\jmath}$. There exist a magnetic field $\vec{B} = -B_0\hat{k}$. Find the motional emf develop across the ends PQ of wire.

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PEM058

4. A rectangular loop with a sliding connector of length $l = 1.0\ m$ is situated in a uniform magnetic field $B = 2T$ perpendicular to the plane of loop. Resistance of connector is $r = 2\Omega$. Two resistances of 6Ω and 3Ω are connected as shown in figure. Find the external force (in N) required to keep the connector moving with a constant velocity $v = 2m/s$.



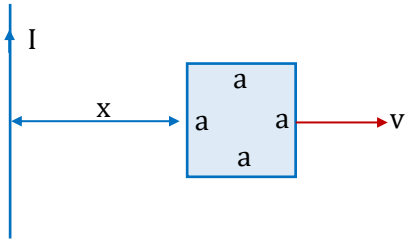
PEM062

5. A uniform magnetic field of magnitude $\frac{1.2}{\pi}\,T$ is in the z-direction. A conducting rod of length 1.5 m lies parallel to the y-axis and oscillates in the x-direction with displacement given by $x = (2\ cm)\cos 120\,\pi t$. What is the emf induced (in volts) in the rod at $t = 1/80$ sec? Round off to nearest integer.

PEM066

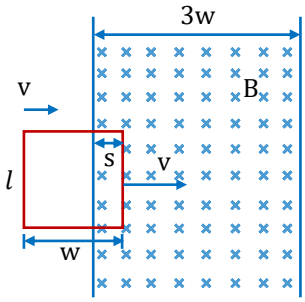
For More Join: @IITJEE_Advanced

6. Now assume that the straight wire carries a current of 50 A and the loop is moved to the right with a constant velocity, $v = 10\ m/s$. Calculate the induced emf (μV) in the loop at the instant when $x = 0.2m$. Take $a = 0.1\ m$ and assume that the loop has a large resistance.



PEM091

7. A rectangular loop of dimensions l *and* w and resistance R moves with constant velocity V to the right as shown in the figure. It continues to move with same speed through a region containing a uniform magnetic field B directed into the plane of the paper and extending a distance 3 w. Sketch the flux, induced emf and external force acting on the loop as a function of the distance.



PEM141

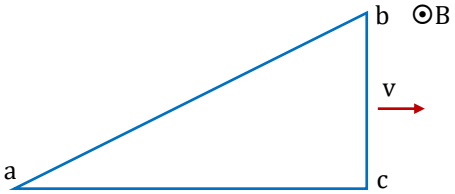
8. A right-angled triangle abc, made from a metallic wire, moves at a uniform speed v in its plane as shown in figure. A uniform magnetic field B exists in the perpendicular direction. Find the emf induced

[a] In the loop abc,

[b] In the segment bc,

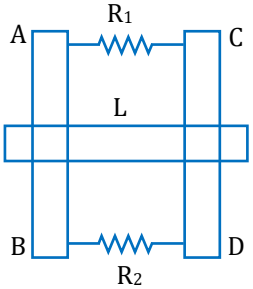
[c] In the segment ac

[d] In the segment ab



PEM143

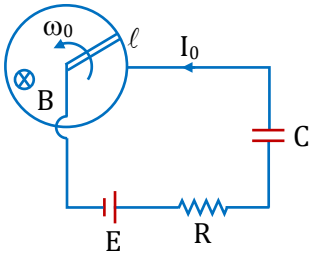
9. Two parallel vertical metallic rails AB *and* CD are separated by 1 m. They are connected at the two ends by resistance R_1 and R_2 as shown in the figure. A horizontally metallic bar L of mass 0.2 kg slides without friction, vertically down the rails under the action of gravity. There is a uniform horizontal magnetic field of $0.6T$ perpendicular to the plane of the rails, it is observed that when the terminal velocity is attained, the power dissipated in R_1 and R_2 are 0.76 W and 1.2 W respectively. Find the terminal velocity of bar L and value R_1 and R_2.



PEM144

For More Join: @IITJEE_Advanced

10. *L* is *s* smooth conducting loop of radius ℓ = 1.0 *m and* fixed in a horizontal plane. A conducting rod of mass *m* = 1.0 *kg* and length slightly greater than ℓ hinged at the centre of the loop can rotate in the horizontal plane such that the free end slides on the rim of the loop. There is a uniform magnetic field of strength *B* = 1.0 *T* directed vertically downward. The rod is rotated with angular velocity ω_0 = 1.0 rad/*s* and left. The fixed end of the rod and the rim of the loop are connected through a battery of e.m.f. *E*, a resistor of resistance *R* = 1.0 Ω, and initially uncharged capacitor of capacitance *C* = 1.0 *F* in series. Find:



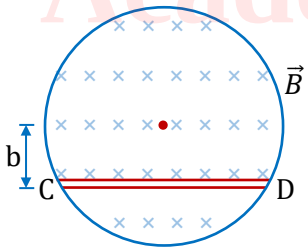
(i) The time dependence of e.m.f. *E* such that the current I_0 = 1.0 *A* in the circuit is constant.

(ii) Energy supplied by the battery by the time rod stops.

PEM160

11. A uniform magnetic field $\vec{B}$ fills a cylindrical volumes of radius *R*. A metal rod *CD* of length *l* is placed inside the cylinder along a chord of the circular cross-section as shown in the figure. If the magnitude of magnetic field increases in the direction of field at a constant rate *dB*/*dt*, find the magnitude and direction of the *EMF* induced in the rod.

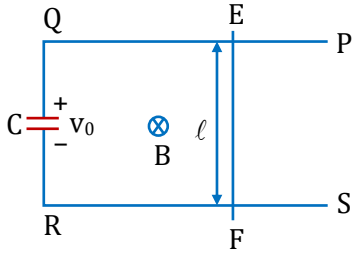
Rankers Academy JEE



PEM082

Self-Inductance and Mutual Inductance

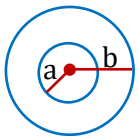
12. In the figure shown '*PQRS*' is a fixed resistanceless conducting frame in a uniform and constant magnetic field of strength *B*. A rod '*EF*' of mass '*m*', length 'ℓ' and resistance *R* can smoothly move on this frame. A capacitor charged to a potential difference 'V_0' initially is connected as shown in the figure. Find the velocity of the rod as function of time '*t*' if it is released at *t* = 0 from rest.



PEM161

For More Join: @IITJEE_Advanced

13. Two concentric and coplanar circular coils have radii a and $b(>>a)$ as shown in figure. Resistance of the inner coil is R. Current in the outer coil is increased from 0 to i, then find the total charge circulating the inner coil.



PEM092

14. A semicircular loop of radius r is placed on the axis of a current carrying circular ring of radius R ($R >> r$), at a distance $R\sqrt{3}$ from the centre of the ring. The plane of the loop is making an angle of 53° with the plane of the ring. If the coefficient of mutual induction between them is $\frac{3\mu_0\pi r^2}{16\times nR}$, then find n:-

PEM031

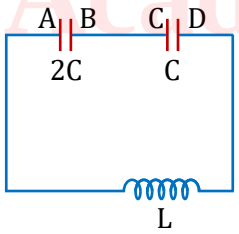
15. An inductor of inductance 2.00 H is joined in series with a resistor of resistance 200 Ω and a battery of emf 2.00 V. At t = 10 ms, find the rate (in mW) at which energy is being stored in magnetic field. (Given e^{-1} = 0.37)

PEM105

L-C Oscillations

16. Two capacitors of capacitances $2C$ and C are connected in series with an inductor of inductance L. Initially capacitors have charge such that $V_B - V_A = 4V_0$ and $V_c - V_D = V_0$. Initial current in the circuit is zero. Find:

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(a) Maximum current that will flow in the circuit.

(b) Potential difference across each capacitor at that instant.

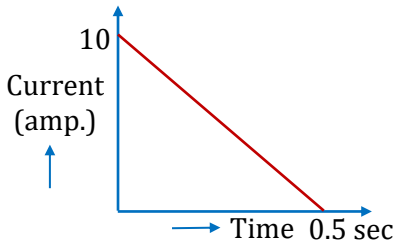
(c) Equation of current flowing towards left in the inductor.

PEM162

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. In a coil of resistance 100 Ω, a current is induced by changing the magnetic flux through it as shown in the figure. The magnitude of change in flux through the coil is :- **[JEE (Main) 2017]**



(1) 250 *Wb* (2) 275 *Wb* (3) 200 *Wb* (4) 225 *Wb*

PEM017

2. A copper wire is wound on a wooden frame, whose shape is that of an equilateral triangle. If the linear dimension of each side of the frame is increased by a factor of 3, keeping the number of turns of the coil per unit length of the frame is the same, then the self inductance of the coil : **[JEE (Main) 2019]**

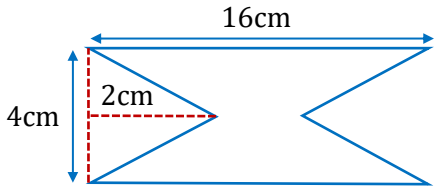
(1) Decreases by a factor of $9\sqrt{3}$ (2) Increases by a factor of 3

(3) Decreases by a factor of 9 (4) Increases by a factor of 27

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PEM018

3. At time $t = 0$ magnetic field of 1000 Gauss is passing perpendicularly through the area defined by the closed loop shown in the figure. If the magnetic field reduces linearly to 500 Gauss, in the next 5*s*, then induced EMF in the loop is : **[JEE (Main) 2020]**



(1) 36 μ*V* (2) 48 μ*V* (3) 56 μ*V* (4) 28 μ*V*

PEM019

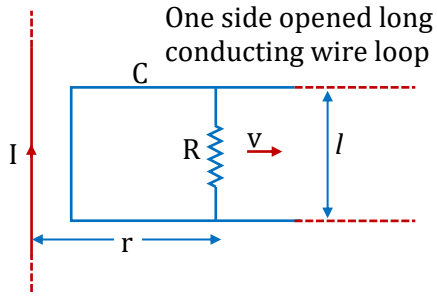
4. A uniform magnetic field *B* exists in a direction perpendicular to the plane of a square loop made of a metal wire. The wire has a diameter of 4 *mm* and a total length of 30 *cm*. The magnetic field changes with time at a steady rate $dB/dt = 0.032\ Ts^{-1}$. The induced current in the loop is close to (Resistivity of the metal wire is $1.23 \times 10^{-8}\ \Omega m$) **[JEE (Main) 2020]**

(1) 0.61 *A* (2) 0.34 *A* (3) 0.43 *A* (4) 0.53 *A*

PEM020

For More Join: @IITJEE_Advanced

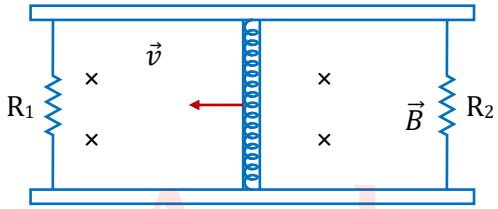
5. An infinitely long straight wire carrying current I, one side opened rectangular loop and a conductor C with a sliding connector are located in the same plane, as shown in the figure. The connector has length l and resistance R. It slides to the right with a velocity v. The resistance of the conductor and the self inductance of the loop are negligible. The induced current in the loop, as a function of separation r, between the connector and the straight wire is : **[JEE (Main) 2020]**



(1) $\frac{\mu_0}{\pi}\frac{Ivl}{Rr}$ (2) $\frac{\mu_0}{2\pi}\frac{Ivl}{Rr}$ (3) $\frac{2\mu_0}{\pi}\frac{Ivl}{Rr}$ (4) $\frac{\mu_0}{4\pi}\frac{Ivl}{Rr}$

PEM021

6. A conducting bar of length L is free to slide on two parallel conducting rails as shown in the figure



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Two resistors R_1 and R_2 are connected across the ends of the rails. There is a uniform magnetic field $\vec{B}$ pointing into the page. An external agent pulls the bar to the left at a constant speed v.

The correct statement about the directions of induced currents I_1 and I_2 flowing through R_1 and R_2 respectively is : **[JEE (Main) 2021]**

(1) Both I_1 and I_2 are in anticlockwise direction

(2) Both I_1 and I_2 are in clockwise direction

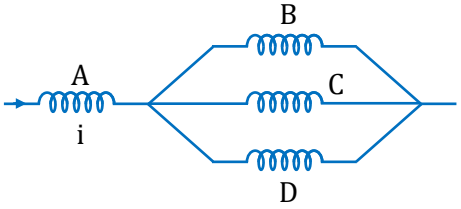
(3) I_1 is in clockwise direction and I_2 is in anticlockwise direction

(4) I_1 is in anticlockwise direction and I_2 is in clockwise direction

PEM163

7. Four identical long solenoids A, B, C and D are connected to each other as shown in the figure. If the magnetic field at the center of A is $3T$, the field at the center of C would be :

(Assume that the magnetic field is confined with in the volume of respective solenoid).



[JEE (Main) 2021]

(1) $12T$ (2) $6T$ (3) $9T$ (4) $1T$

PEM164

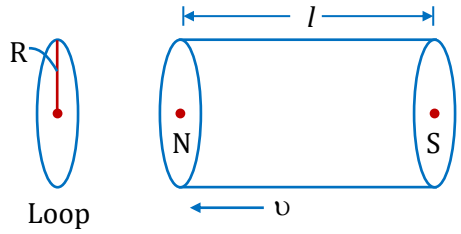
For More Join: @IITJEE_Advanced

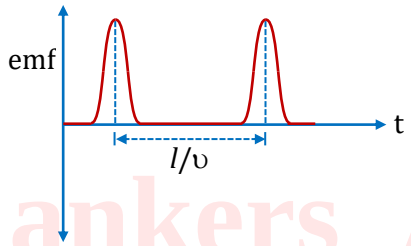
8. In a coil of resistance 8Ω, the magnetic flux due to an external magnetic field varies with time as $\phi=\frac{2}{3}\left(9-t^2\right)$. The value of total heat produced in the coil, till the flux becomes zero, will be_____*J*.

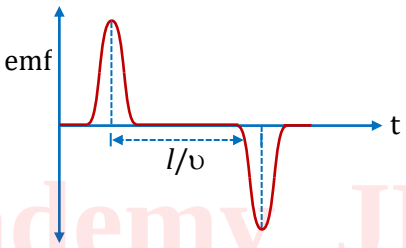
[JEE (Main) 2022]

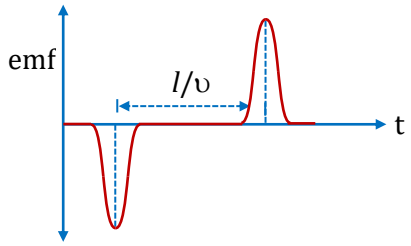
PEM165

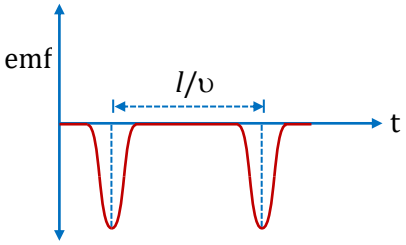
9. A bar magnet is passing through a conducting loop of radius *R* with velocity υ. The radius of the bar magnet is such that it just passes through the loop. The induced e.m.f. in the loop can be represented by the approximate curve : **[JEE (Main) 2022]**



(1)


(2)


(3)


(4)


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PEM166

10. A square shaped coil of area 70 cm^2 having 600 turns rotates in a magnetic field of 0.4 wbm^{-2}, about an axis which is parallel to one of the side of the coil and perpendicular to the direction of field. If the coil completes 500 revolution in a minute, the instantaneous emf when the plane of the coil is inclined at 60° with the field, will be ___ V. (Take $\pi=\frac{22}{7}$) **[JEE (Main) 2023]**

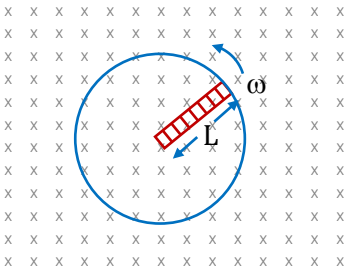
PEM146

11. An oscillating LC circuit consists of a 75 *mH* inductor and a 1.2μF capacitor. If the maximum charge to the capacitor is 2.7μC. The maximum current in the circuit will be ___*mA*. **[JEE (Main) 2023]**

PEM147

For More Join: @IITJEE_Advanced

12. A metallic rod of length 'L' is rotated with an angular speed of 'ω' normal to a uniform magnetic field 'B' about an axis passing through one end of rod as shown in figure. The induced emf will be :



[JEE (Main) 2023]

(1) $\frac{1}{4}B^2L\omega$ (2) $\frac{1}{4}BL^2\omega$ (3) $\frac{1}{2}BL^2\omega$ (4) $\frac{1}{2}B^2L^2\omega$

PEM167

13. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: A bar magnet dropped through a metallic cylindrical pipe takes more time to come down compared to a non–magnetic bar with same geometry and mass.

Reason R: For the magnetic bar, Eddy currents are produced in the metallic pipe which oppose the motion of the magnetic bar.

In the light of the above statements, choose the correct answer from the options given below:

[JEE (Main) 2023]

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(1) Both **A** and ***R*** are true but ***R*** is NOT the correct explanation of **A**

(2) **A** is true but ***R*** is false

(3) Both **A** and ***R*** are true and ***R*** is the correct explanation of **A**

(4) **A** is false but ***R*** is true

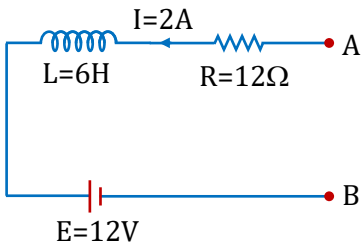
PEM168

14. A coil has an inductance of $2H$ and resistance of 4Ω. A $10V$ is applied across the coil. The energy stored in the magnetic field after the current has built up to its equilibrium value will be _____$\times 10^{-2}J$.

[JEE (Main) 2023]

PEM169

15. As per the given figure, if $\frac{dI}{dt} = -1$ A/s then the value of V_{AB} at this instant will be _____ V.



[JEE (Main) 2023]

PEM170

For More Join: @IITJEE_Advanced

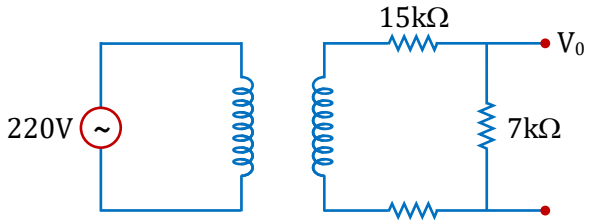
16. A rectangular loop of sides 12 *cm* and 5 *cm*, with its sides parallel to the *x*-axis and *y*-axis respectively moves with a velocity of 5 *cm*/*s* in the positive *x* axis direction, in a space containing a variable magnetic field in the positive *z* direction. The field has a gradient of 10^{-3} *T*/*cm* along the negative x direction and it is decreasing with time at the rate of 10^{-3} *T*/*s*. If the resistance of the loop is 6 *m*Ω, the power dissipated by the loop as heat is _____ × 10^{-9} *W*. **[JEE (Main) 2024]**

PEM171

17. A small square loop of wire of side ℓ is placed inside a large square loop of wire of side L ($L = \ell^2$). The loops are coplanar and their centers coinside. The value of the mutual inductance of the system is $\sqrt{x} \times 10^{-7}$ H, where x = _____. **[JEE (Main) 2024]**

PEM172

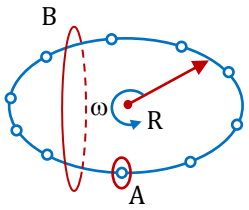
18. Primary coil of a transformer is connected to 220 *V* ac. Primary and secondary turns of the transforms are 100 and 10 respectively. Secondary coil of transformer is connected to two series resistance shown in shown in figure. The output voltage (V_0) is : **[JEE (Main) 2024]**



(1) 7 *V* (2) 15 *V* (3) 44 *V* (4) 22 *V*

PEM175

19. *N* equally spaced charges each of value *q*, are placed on a circle of radius *R*. The circle rotates about its axis with an angular velocity ω as shown in the figure. A bigger Amperian loop *B* encloses the whole circle where as a smaller Amperian loop *A* encloses a small segment. The difference between enclosed currents, $I_A - I_B$, for the given Amperian loops is **[JEE (Main) 2025]**



(1) $\frac{N^2}{2\pi}q\omega$ (2) $\frac{2\pi}{N}q\omega$ (3) $\frac{N}{2\pi}q\omega$ (4) $\frac{N}{\pi}q\omega$

PEM213

20. Consider I_1 and I_2 are the currents flowing simultaneously in two nearby coils 1 and 2, respectively. If L_1 = self inductance of coil 1, M_{12} = mutual inductance of coil 1 with respect to coil 2, then the value of induced emf in coil 1 will be **[JEE (Main) 2025]**

(1) $\varepsilon_1 = -L_1\frac{dI_1}{dt} + M_{12}\frac{dI_2}{dt}$ (2) $\varepsilon_1 = -L_1\frac{dI_1}{dt} - M_{12}\frac{dI_1}{dt}$

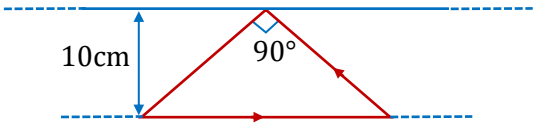
(3) $\varepsilon_1 = -L_1\frac{dI_1}{dt} - M_{12}\frac{dI_2}{dt}$ (4) $\varepsilon_1 = -L_1\frac{dI_2}{dt} - M_{12}\frac{dI_1}{dt}$

PEM214

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. A conducting loop in the shape of right-angled isosceles triangle of height 10 *cm* is kept such that the 90° vertex is very close to an infinitely long conducting wire (see the figure). The wire is electrically insulated from the loop. The hypotenuse of the triangle is parallel to the wire. The current in the triangular loop is in counterclockwise direction and increased at constant rate of 10 $A\ s^{-1}$. Which of the following statement(s) is(are) true? **[JEE (Advanced) 2016]**



(A) The induced current in the wire is in opposite direction to the current along the hypotenuse.

(B) There is a repulsive force between the wire and the loop.

(C) If the loop is rotated at a constant angular speed about the wire, an additional emf of $\left(\frac{\mu_0}{\pi}\right)$ volt is induced in the wire.

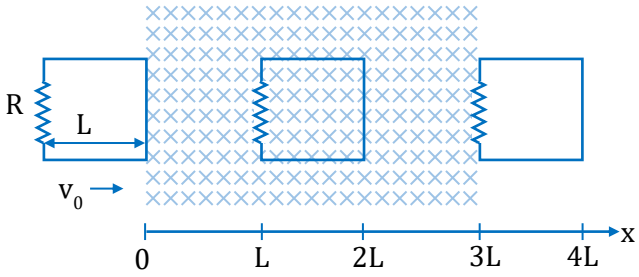
(D) The magnitude of induced emf in the wire is $\left(\frac{\mu_0}{\pi}\right)$ volt.

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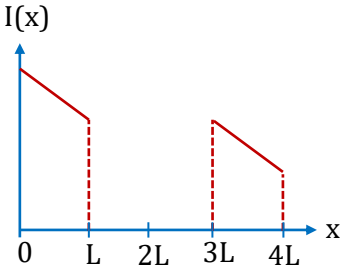
PEM034

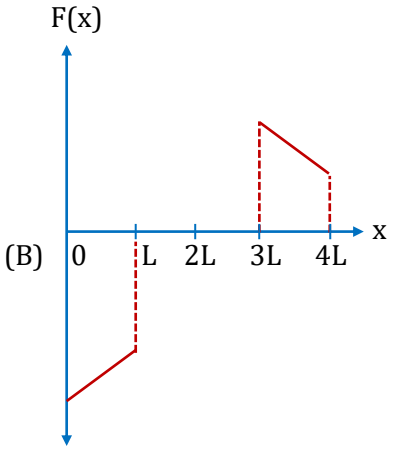
2. A rigid wire loop of square shape having side of length L and resistance R is moving along the x-axis with a constant velocity v_0 in the plane of the paper. At $t = 0$, the right edge of the loop enters a region of length $3L$ where there is a uniform magnetic field B_0 into the plane of the paper, as shown in the figure. For sufficiently large v_0, the loop eventually crosses the region. Let x be the location of the right edge of the loop. Let $v(x)$, $I(x)$ and $F(x)$ represent the velocity of the loop, current in the loop, and force on the loop, respectively, as a function of x. Counter-clockwise current is taken as positive. Which of the following schematic plot(s) is(are) correct? (Ignore gravity)

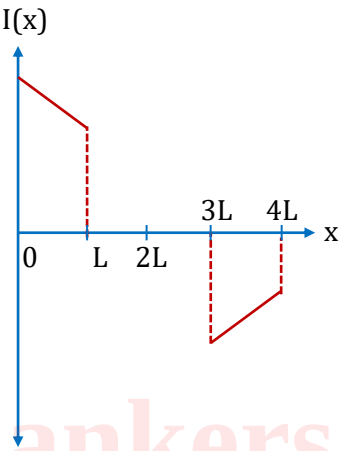
[JEE (Advanced) 2016]

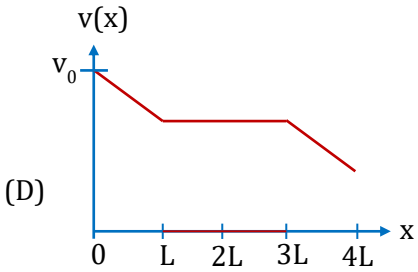


For More Join: @IITJEE_Advanced

(A)


(B)


(C)


(D)


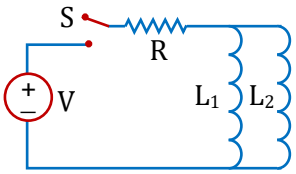
Rankers Academy JEE

PEM035

3. Two inductors L_1 (inductance 1 mH, internal resistance 3 Ω) and L_2 (inductance 2mH, internal resistance 4Ω), and a resistor R (resistance 12 Ω) are all connected in parallel across a 5V battery. The circuit is switched on at time $t = 0$. The ratio of the maximum to the minimum current (I_{max}/I_{min}) drawn from the battery is. **[JEE (Advanced) 2016]**

PEM176

4. A source of constant voltage V is connected to a resistance R and two ideal inductors L_1 and L_2 through a switch S as shown. There is no mutual inductance between the two inductors. The switch S is initially open. At $t = 0$, the switch is closed and current begins to flow. Which of the following options is/are correct? **[JEE (Advanced) 2017]**



(A) The ratio of the currents through L_1 and L_2 is fixed at all times ($t > 0$).

(B) After a long time, the current through L_1 will be $\frac{V}{R}\frac{L_2}{L_1+L_2}$.

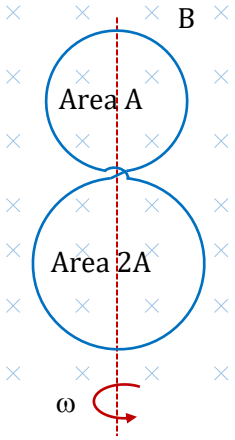
(C) After a long time, the current through L_2 will be $\frac{V}{R}\frac{L_1}{L_1+L_2}$.

(D) At $t = 0$, the current through the resistance R is $\frac{V}{R}$.

PEM177

For More Join: @IITJEE_Advanced

5. A circular insulated copper wire loop is twisted to form two loops of area A and $2A$ as shown in the figure. At the point of crossing the wires remain electrically insulated from each other. The entire loop lies in the plane (of the paper). A uniform magnetic field $\vec{B}$ points into the plane of the paper. At $t = 0$, the loop starts rotating about the common diameter as axis with a constant angular velocity ω in the magnetic field. Which of the following options is/are correct? **[JEE (Advanced) 2017]**



(A) The rate of change of the flux is maximum when the plane of the loops is perpendicular to plane of the paper.

(B) The net emf induced due to both the loops is proportional to cos ωt.

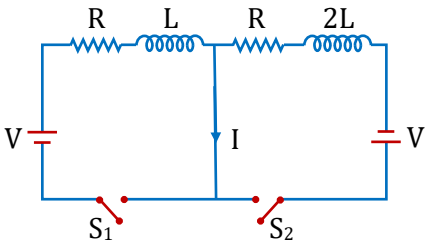
(C) The emf induced in the loop is proportional to the sum of the areas of the two loops.

(D) The amplitude of the maximum net emf induced due to both the loops is equal to the amplitude of maximum emf induced in the smaller loop alone.

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PEM036

6. In the figure below, the switches S_1 and S_2 are closed simultaneously at $t = 0$ and a current starts to flow in the circuit. Both the batteries have the same magnitude of the electromotive force (*emf*) and the polarities are as indicated in the figure. Ignore mutual inductance between the inductors. The current I in the middle wire reaches its maximum magnitude I_{max} at time $t = \tau$. Which of the following statement(s) is (are) true?



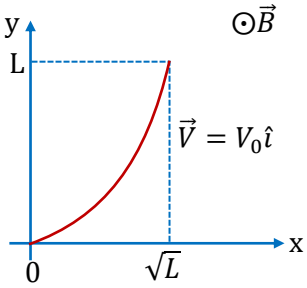
[JEE (Advanced) 2018]

(A) $I_{max} = \frac{V}{2R}$ (B) $I_{max} = \frac{V}{4R}$ (C) $\tau = \frac{L}{R}\ell n2$ (D) $\tau = \frac{2L}{R}\ell n2$

PEM178

7. A conducting wire of parabolic shape, initially $y = x^2$, is moving with velocity $\vec{V} = V_0\hat{i}$ in a non-uniform magnetic field $\vec{B} = B_0\left(1+\left(\frac{y}{L}\right)^{\beta}\right)\hat{k}$, as shown in figure. If V_0, B_0, L and b are positive constants and $\Delta\phi$ is the potential difference developed between the ends of the wire, then the correct statement(s) is/are: **[JEE (Advanced) 2019]**

For More Join: @IITJEE_Advanced



(A) $|\Delta\phi|$ remains the same if the parabolic wire is replaced by a straight wire, $y = x$ initially, of length $\sqrt{2}L$.

(B) $|\Delta\phi|$ is proportional to the length of the wire projected on the y-axis.

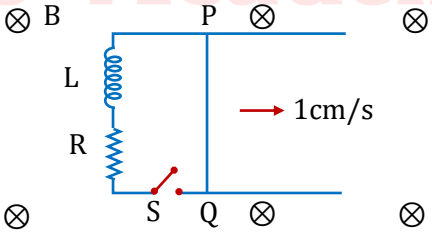
(C) $|\Delta\phi| = \frac{1}{2}B_0V_0L$ for $b = 0$.

(D) $|\Delta\phi| = \frac{4}{3}B_0V_0L$ for $b = 2$.

PEM037

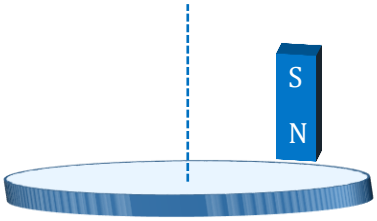
8. A 10 *cm* long perfectly conducting wire PQ is moving with a velocity 1cm/s on a pair of horizontal rails of zero resistance. One side of the rails is connected to an inductor $L = 1\ mH$ and a resistance $R = 1\Omega$ as shown in figure. The horizontal rails, L and R lie in the same plane with a uniform magnetic field $B = 1T$ perpendicular to the plane. If the key S is closed at certain instant, the current in the circuit after 1 millisecond is $x \times 10^{-3}A$, where the value of x is_____.
[Assume the velocity of wire PQ remains constant (1 cm/s) after key S is closed.
Given : $e^{-1} = 0.37$, where e is base of the natural logarithm] **[JEE (Advanced) 2019]**

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PEM038

9. A light disc made of aluminum (a nonmagnetic material) is kept horizontally and is free to rotate about its axis as shown in the figure. A strong magnet is held vertically at a point above the disc away from its axis. On revolving the magnet about the axis of the disc, the disc will (figure is schematic and not drawn to scale)-



[JEE (Advanced) 2020]

(A) rotate in the direction opposite to the direction of magnet's motion

(B) rotate in the same direction as the direction of magnet's motion

(C) not rotate and its temperature will remain unchanged

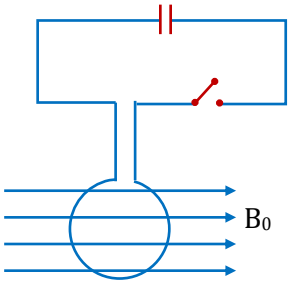
(D) not rotate but its temperature will slowly rise

PEM039

For More Join: @IITJEE_Advanced

10. A circular coil of radius R and N turns has negligible resistance. As shown in the schematic figure, its two ends are connected to two wires and it is hanging by those wires with its plane being vertical. The wires are connected to a capacitor with charge Q through a switch. The coil is in a horizontal uniform magnetic field B_0 parallel to the plane of the coil. When the switch is closed, the capacitor gets discharged through the coil in a very short time. By the time the capacitor is discharged fully, magnitude of the angular momentum gained by the coil will be (assume that the discharge time is so short that the coil has hardly rotated during this time)-

[JEE (Advanced) 2020]

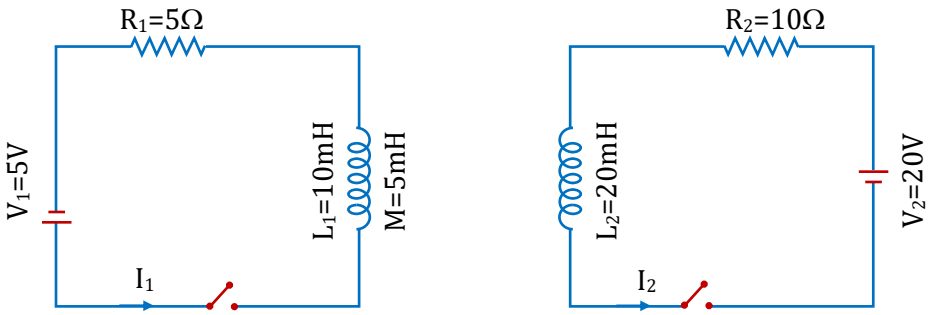


(A) $\frac{\pi}{2}NQB_oR^2$ (B) πNQB_0R^2 (C) $2\pi NQB_0R^2$ (D) $4\pi NQB_0R^2$

PEM040

11. The inductors of two LR circuits are placed next to each other, as shown in the figure. The values of the self-inductance of the inductors, resistances, mutual-inductance and applied voltages are specified in the given circuit. After both the switches are closed simultaneously, the total work done by the batteries against the induced EMF in the inductors by the time the currents reach their steady state values is_____ mJ.

[JEE (Advanced) 2020]



PEM179

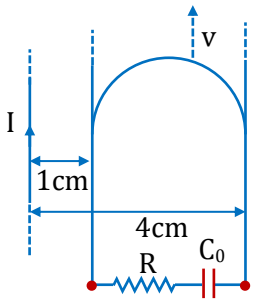
12. A long straight wire carries a current, $I = 2$ ampere. A semi-circular conducting rod is placed beside it on two conducting parallel rails of negligible resistance. Both the rails are parallel to the wire. The wire, the rod and the rails lie in the same horizontal plane, as shown in the figure. Two ends of the semi-circular rod are at distances $1cm$ and $4\ cm$ from the wire. At time $t = 0$, the rod starts moving on the rails with a speed $v = 3.0\ m/s$ (see the figure).

A resistor $R = 1.4\ \Omega$ and a capacitor $C_0 = 5.0\ \mu F$ are connected in series between the rails. At time $t = 0$, C_0 is uncharged. Which of the following statement(s) is(are) correct ?

[$\mu_0 = 4\pi \times 10^{-7}$ SI units. Take ln 2 = 0.7]

[JEE (Advanced) 2021]

For More Join: @IITJEE_Advanced



(A) Maximum current through R is 1.2×10^{-6} ampere.

(B) Maximum current through R is 3.8×10^{-6} ampere.

(C) Maximum charge on capacitor C_0 is 8.4×10^{-12} coulomb.

(D) Maximum charge on capacitor C_0 is 2.4×10^{-12} coulomb.

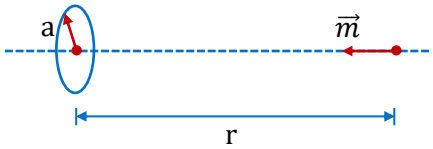
PEM180

QUESTION STEM FOR QUESTION NOS. 13 AND 14

A special metal S conducts electricity without any resistance. A closed wire loop, made of S, does not allow any change in flux through itself by inducing a suitable current to generate a compensating flux. The induced current in the loop cannot decay due to its zero resistance. This current gives rise to a magnetic moment which in turn repels the source of magnetic field or flux. Consider such a loop, of radius a, with its center at the origin. A magnetic dipole of moment m is brought along the axis of this loop from infinity to a point at distance r (>> a) from the center of the loop with its north pole always facing the loop, as shown in the figure below.

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The magnitude of magnetic field of a dipole m, at a point on its axis at distance r, is $\frac{\mu_0}{2\pi}\frac{m}{r^3}$, where μ_0 is the permeability of free space. The magnitude of the force between two magnetic dipoles with moments, m_1 and m_2, separated by a distance r on the common axis, with their north poles facing each other, is $\frac{km_1m_2}{r^4}$, where k is a constant of appropriate dimensions. The direction of this force is along the line joining the two dipoles. **[JEE (Advanced) 2021]**



13. When the dipole m is placed at a distance r from the center of the loop (as shown in the figure), the current induced in the loop will be proportional to :

(A) $\frac{m}{r^3}$ (B) $\frac{m^2}{r^2}$ (C) $\frac{m}{r^2}$ (D) $\frac{m^2}{r}$

PEM041

14. The work done in bringing the dipole from infinity to a distance r from the center of the loop by the given process is proportional to :

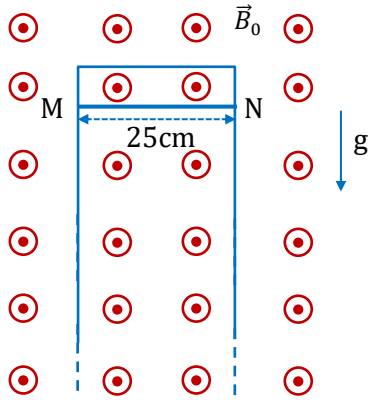
(A) $\frac{m}{r^5}$ (B) $\frac{m^2}{r^5}$ (C) $\frac{m^2}{r^6}$ (D) $\frac{m^2}{r^7}$

PEM042

For More Join: @IITJEE_Advanced

15. A thin conducting rod MN of mass 20 gm, length 25 cm and resistance 10 Ω is held on frictionless, long, perfectly conducting vertical rails as shown in the figure. There is a uniform magnetic field $B_0 = 4T$ directed perpendicular to the plane of the rod-rail arrangement. The rod is released from rest at time $t = 0$ and it moves down along the rails. Assume air drag is negligible. Match each quantity in List-I with an appropriate value from List-II, and choose the correct option.

[Given: The acceleration due to gravity $g = 10\ ms^{-2}$ and $e^{-1} = 0.4$]



	List-I		List-II
(P)	At $t = 0.2s$, the magnitude of the induced emf in Volt	(1)	0.07
(Q)	At $t = 0.2s$, the magnitude of the magnetic force in Newton	(2)	0.14
(R)	At $t = 0.2s$, the power dissipated as heat in Watt	(3)	1.20
(S)	The magnitude of terminal velocity of the rod in ms^{-1}	(4)	0.12
		(5)	2.00

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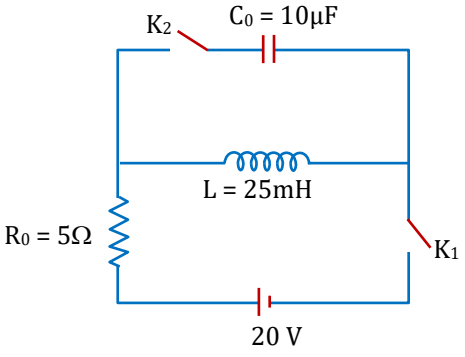
[JEE (Advanced) 2023]

(A) $P \to 5, Q \to 2, R \to 3, S \to 1$ (B) $P \to 3, Q \to 1, R \to 4, S \to 5$

(C) $P \to 4, Q \to 3, R \to 1, S \to 2$ (D) $P \to 3, Q \to 4, R \to 2, S \to 5$

PEM155

16. The circuit shown in the figure contains an inductor L, a capacitor C_0, a resistor R_0 and an ideal battery. The circuit also contains two keys K_1 and K_2. Initially, both the keys are open and there is no charge on the capacitor. At an instant, key K_1 is closed and immediately after this the current in R_0 is found to be I_1. After a long time, the current attains a steady state value I_2. Thereafter, K_2 is closed and simultaneously K_1 is opened and the voltage across C_0 oscillates with amplitude V_0 and angular frequency ω_0.



For More Join: @IITJEE_Advanced

Match the quantities mentioned in List-I with their values in List-II and choose the correct option.

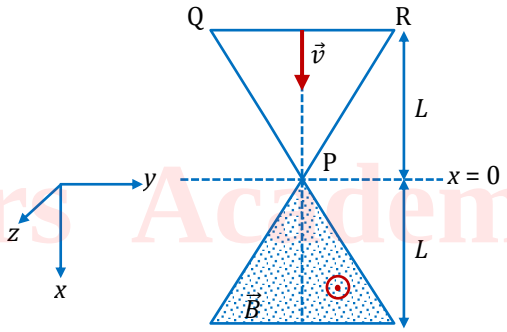
	List-I		List-II
(P)	The value of I_1 in Ampere is	(1)	0
(Q)	The value of I_2 in Ampere is	(2)	2
(R)	The value of ω_0 in kilo-radians/s is	(3)	4
(S)	The value of V_0 in Volt is	(4)	20
		(5)	200

[JEE (Advanced) 2024]

(A) P → 1; Q → 3; R → 2; S → 5
(B) P → 1; Q → 2; R → 3; S → 5
(C) P → 1; Q → 3; R → 2; S → 4
(D) P → 2; Q → 5; R → 3; S → 4

PEM181

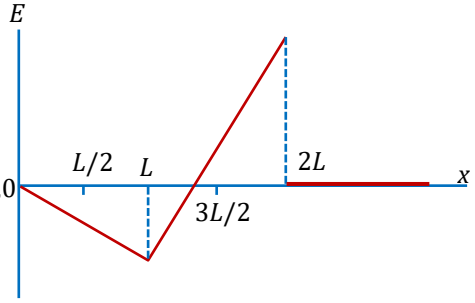
17. A region in the form of an equilateral triangle (in *x*-*y* plane) of height L has a uniform magnetic field $\vec{B}$ pointing in the +z-direction. A conducting loop *PQR*, in the form of an equilateral triangle of the same height L, is placed in the *x*-*y* plane with its vertex *P* at $x = 0$ in the orientation shown in the figure. At $t = 0$, the loop starts entering the region of the magnetic field with a uniform velocity $\vec{v}$ along the +*x*-direction. The plane of the loop and its orientation remain unchanged throughout its motion.

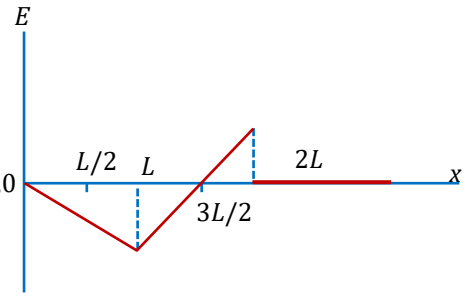


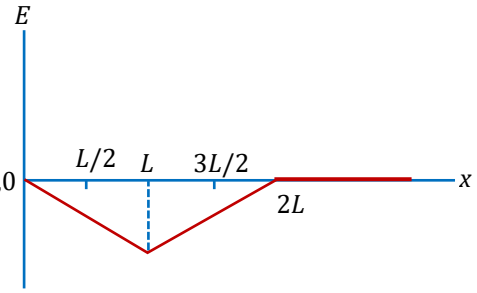
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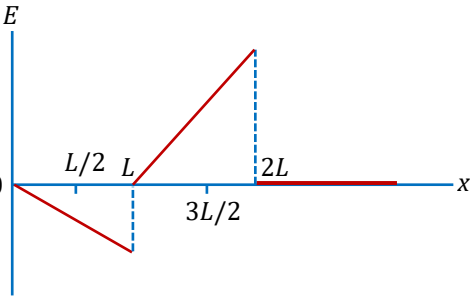
Which of the following graph best depicts the variation of the induced emf (E) in the loop as a function of the distance (x) starting from $x = 0$?

[JEE (Advanced) 2024]

(A)


(B)


(C)


(D)


PEM182

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	D	C	C	B	D	D	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	B	A	A	C	B	A	A	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	A	C	A	B	B	D	B	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	A	B	A	A	D	B	A	B
Que.	41	42	43	44						
Ans.	A	A	B	D						

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AC	ABCD	AC	ABC	BD	AB	BC	ACD	ABD	BD
Que.	11	12	13	14	15	16	17	18	19	
Ans.	AC	C	B	D	C	D	A	C	B	

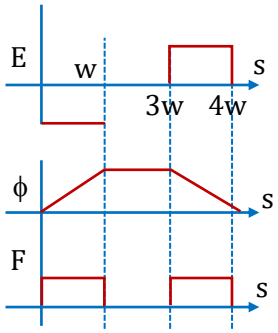
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EXERCISE (S)

1. **Ans.** 6.5 V **2. Ans.** 32 **3. Ans.** $\lambda V_y B_0$ **4. Ans.** 2 **5. Ans.** 4

6. **Ans.** $1.7 \times 10^{-5}\ V$

7. **Ans.**



8. **Ans.** (a) zero, (b) $vB(bc)$, positive at c, (c) zero (d) $vB(bc)$, positive at a

9. **Ans.** $V = 1\ ms^{-1}$, $R_1 = 0.5\ \Omega$, $R_2 = 0.30\ \Omega$

10. **Ans.** (i) $\frac{1}{2}+\frac{7t}{4}$ (ii) $\frac{13}{18} J$ J

11. Ans. $\left(\frac{1}{2}\frac{dB}{dt}\sqrt{R^2-\frac{l^2}{4}}\right)$

12. **Ans.** $v = \frac{B\ell C\ V_0}{m+B^2\ell^2 C}\left(1 - e^{-\left(\frac{B^2\ell^2}{mR}+\frac{1}{RC}\right)t}\right)$

13. Ans. $\left(\frac{\mu_0 i a^2 \pi}{2Rb}\right)$

14. **Ans.** 10

15. Ans. 4.6

16. **Ans.** (a) $I_{max} = \left(\sqrt{\frac{6C}{L}}\right) v_0$; (b) $3\,v_0$, $3\,v_o$; (c) $i = I_{max}\sin\omega t$; $\omega = \left(\sqrt{\frac{3}{2LC}}\right)$

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	3	1	2	3	4	2	3	44
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	9	3	3	625	30	216	128	1	3	3

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	BD	CD	8	ABC	AD	BD	ABD	0.63	B	B
Que.	11	12	13	14	15	16	17			
Ans.	55	AC	A	C	D	A	A			

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